

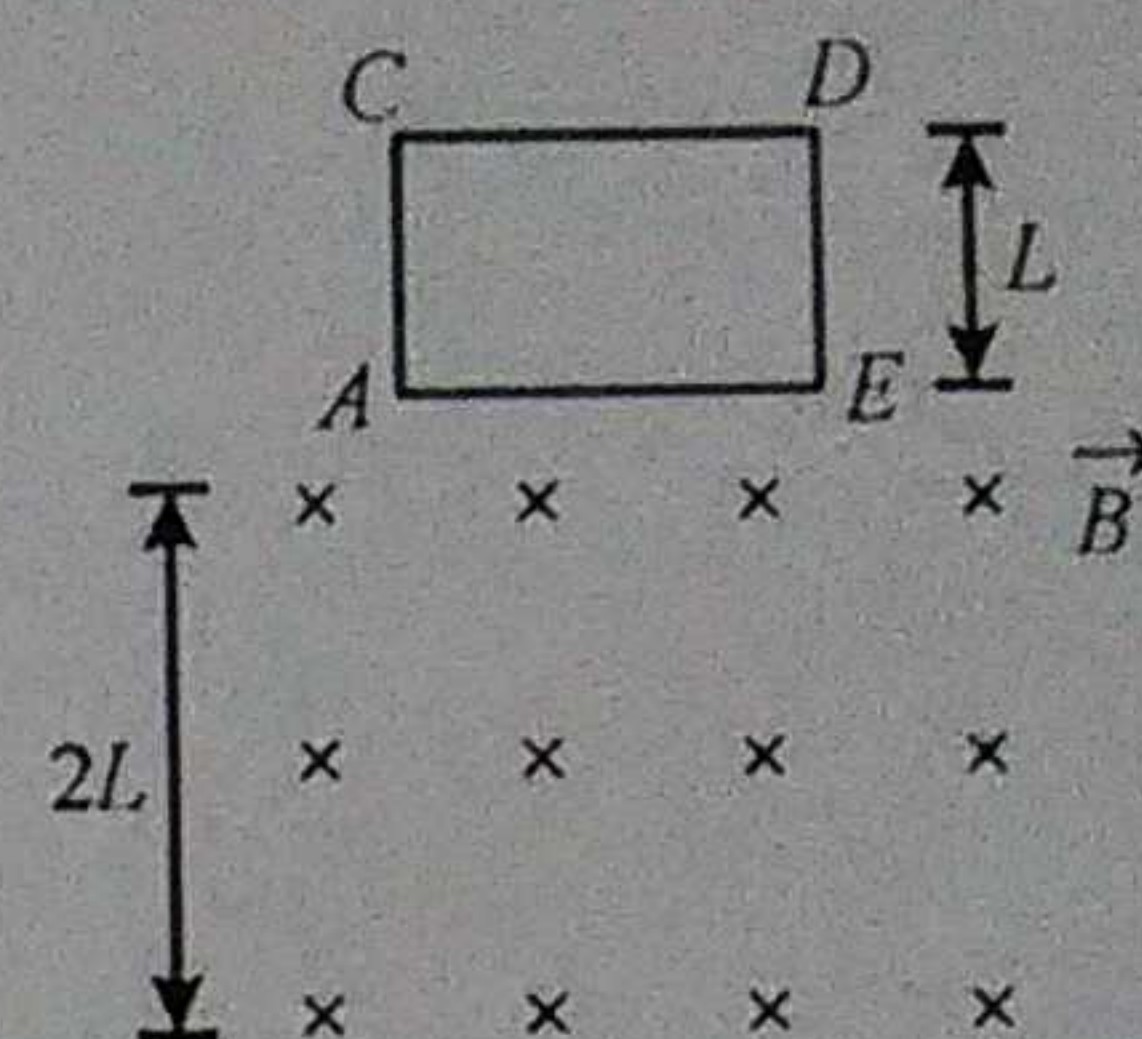
# MAGNETICS & Electromagnetic Induction

Type of CPP - Objective & Subjective

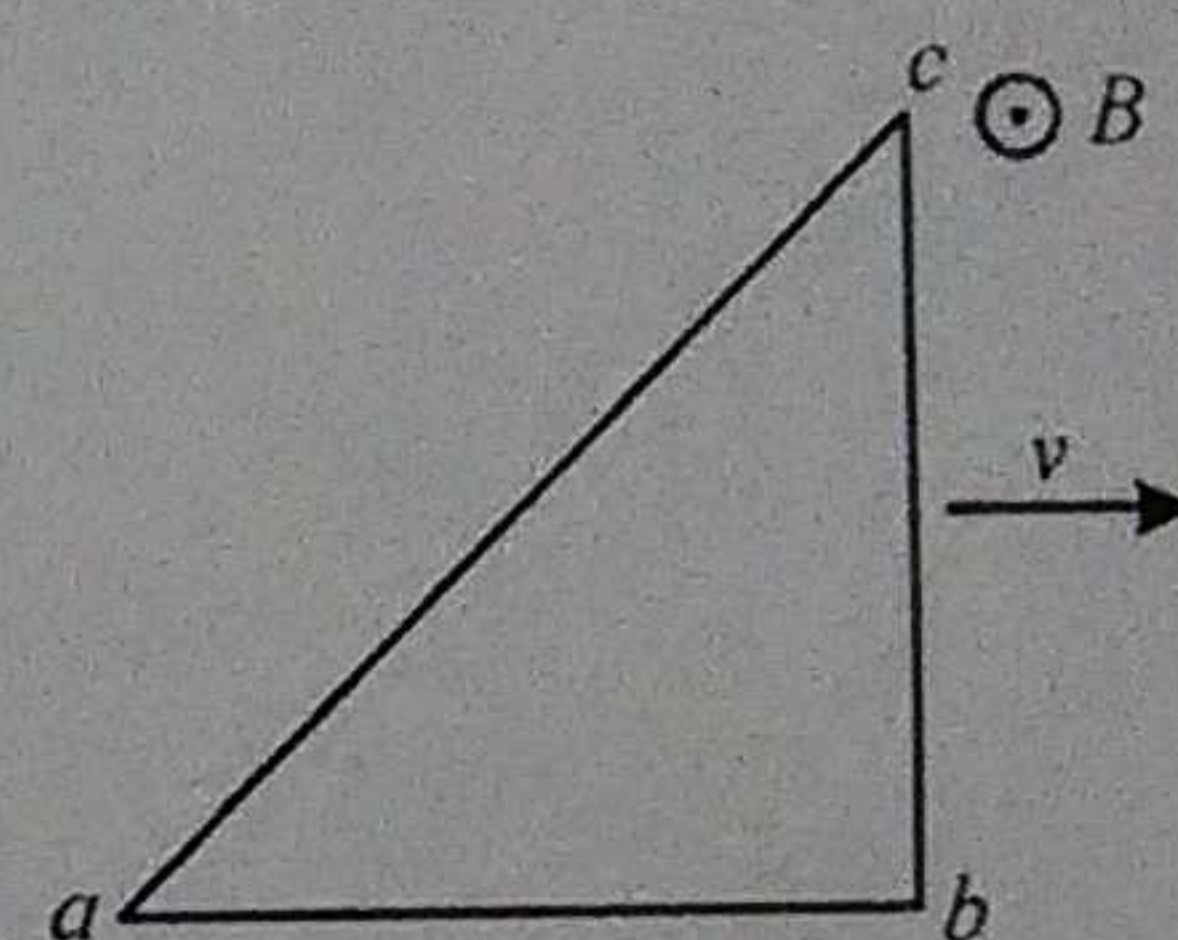
Level - MODERATE

1. A wire-loop confined in a plane is rotated in its own plane with some angular velocity. A uniform magnetic field exist in the region. Find the emf induced in the loop.

2. A square coil  $ACDE$  with its plane vertical is released from rest in a horizontal uniform magnetic field  $\vec{B}$  of length  $2L$ . The acceleration of the coil is :
- (A) Less than  $g$  for all the time till the loop crosses the magnetic field completely  
 (B) Less than  $g$  when it enters the field and greater than  $g$  when it comes out of the field  
 (C)  $g$  all the time  
 (D) Less than  $g$  when it enters and comes out of the field but equal to  $g$  when it is within the field

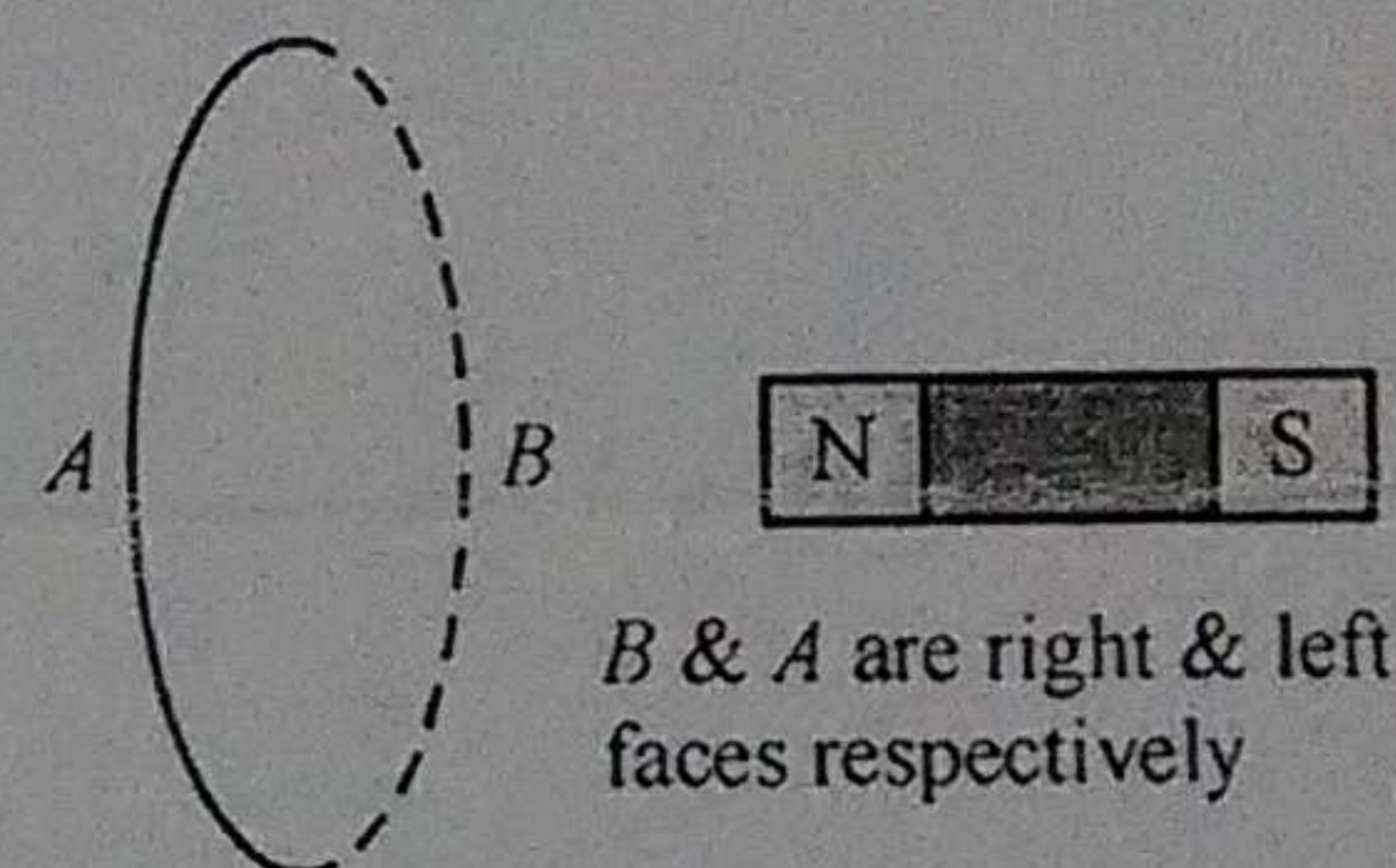


3. A right angled triangle  $abc$ , made from a metallic wire, moves at a uniform speed  $v$  in its plane as shown in the figure. A uniform magnetic field  $B$  exists in the perpendicular direction. Find the emf induced (a) in the loop  $abc$ , (b) in the segment  $bc$ , (c) in the segment  $ac$  and (d) in the segment  $ab$ .

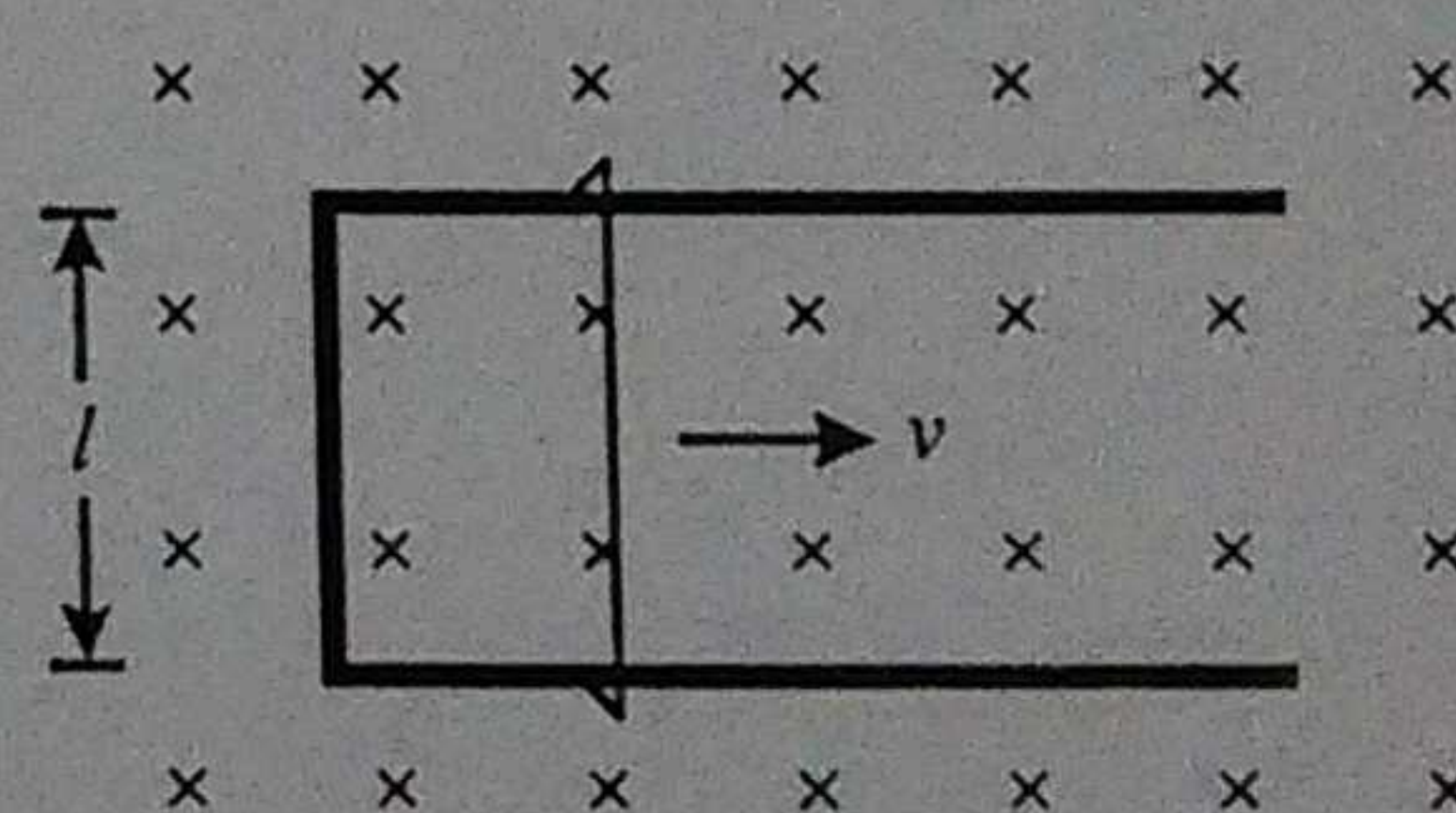


4. In the figure shown, the magnet is pushed towards the fixed ring along the axis of the ring and it passes through the ring :

- (A) When magnet goes towards the ring the face  $B$  becomes south pole and the face  $A$  becomes north pole  
 (B) When magnet goes away from the ring the face  $B$  becomes north pole and the face  $A$  becomes south pole  
 (C) When magnet goes away from the ring the face  $A$  becomes north pole and the face  $B$  becomes south pole  
 (D) The face  $A$  will always be a north pole.

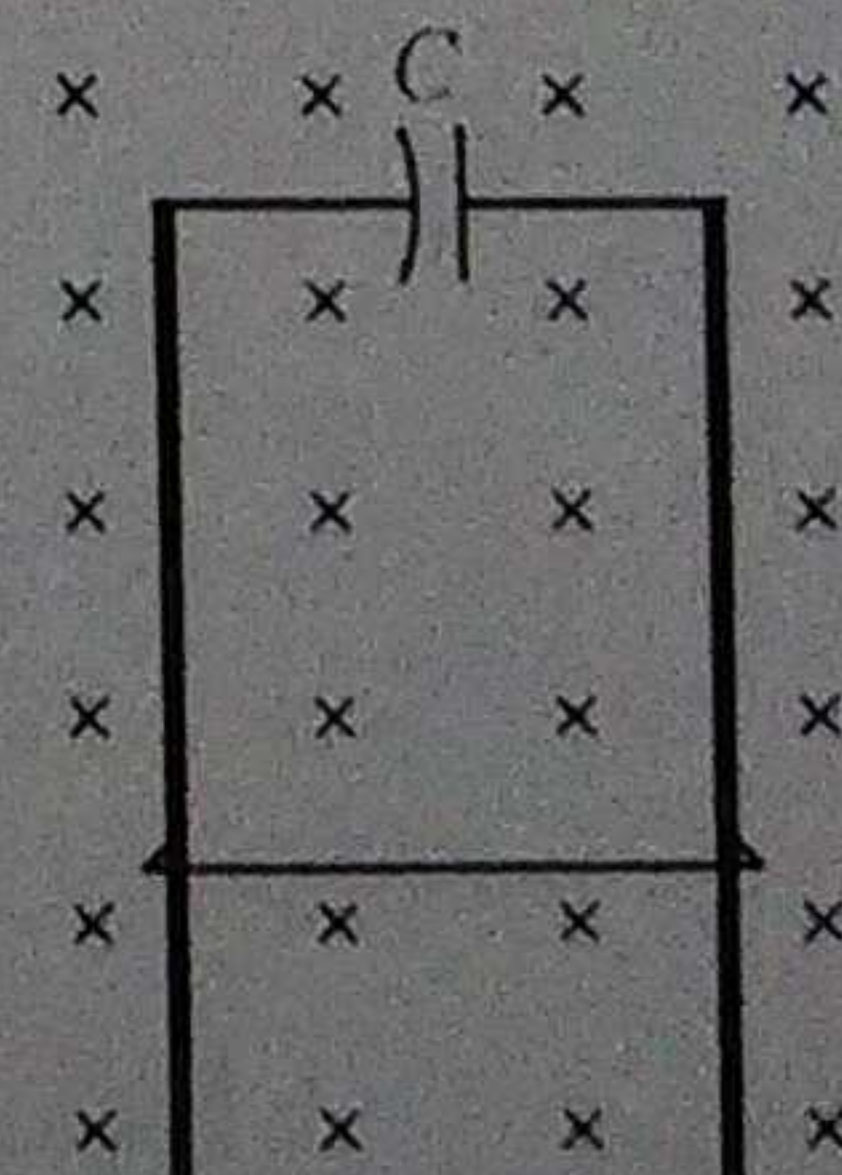


5. Figure shows a long U-shaped wire of width  $l$  placed in a perpendicular magnetic field  $B$ . A wire of length  $l$  is slid on the U-shaped wire with a constant velocity  $v$  towards right. The resistance of all the wires is  $r$  per unit length. At  $t = 0$ , the sliding wire is close to the left edge of the U-shaped wire. Draw an equivalent circuit diagram at time  $t$ , showing the induced emf as a battery. Calculate the current in the circuit.



6. Consider the situation of the previous problem. (a) Calculate the force needed to keep the sliding wire moving with a constant velocity  $v$ . (b) If the force needed just after  $t = 0$  is  $F_0$ , find the time at which the force needed will be  $F_0/2$ .

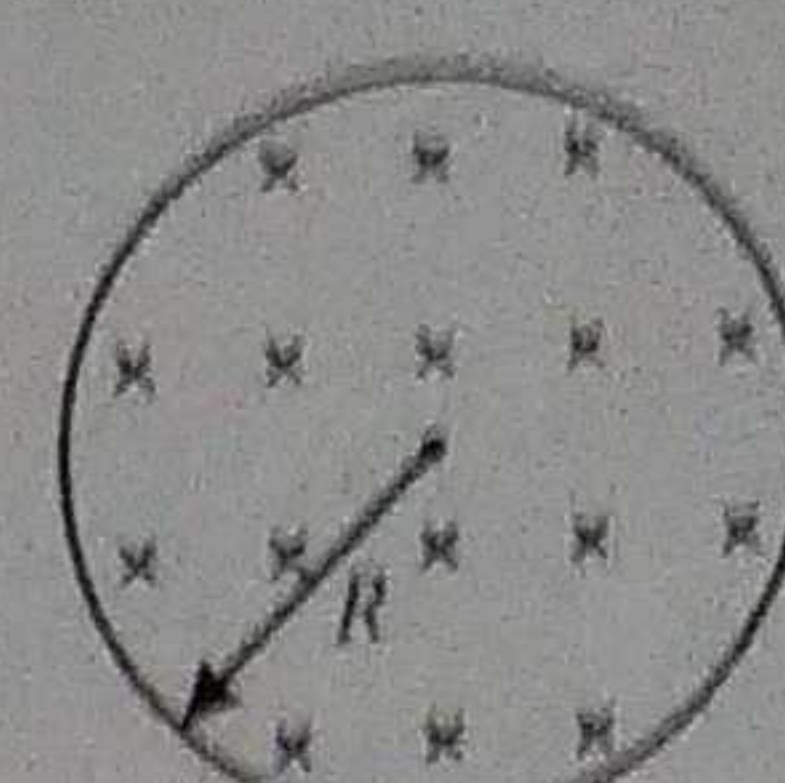
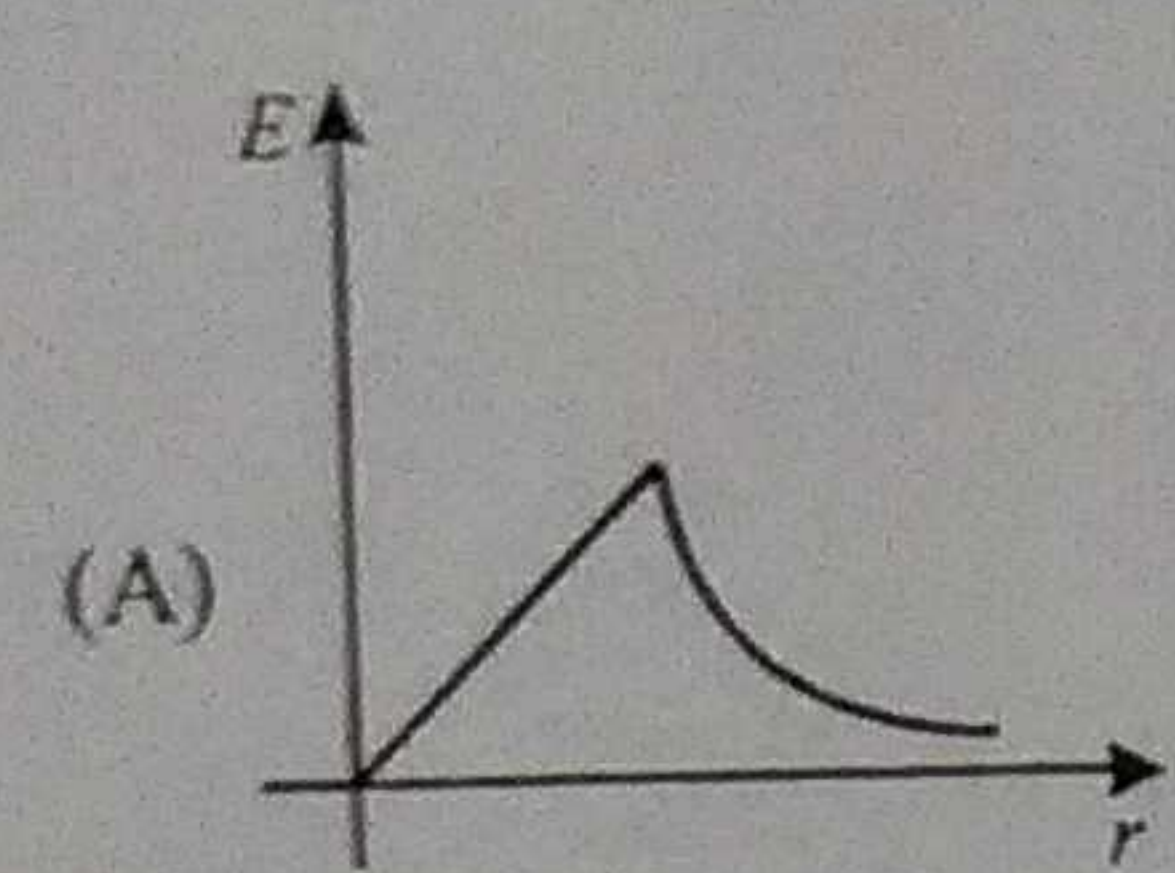
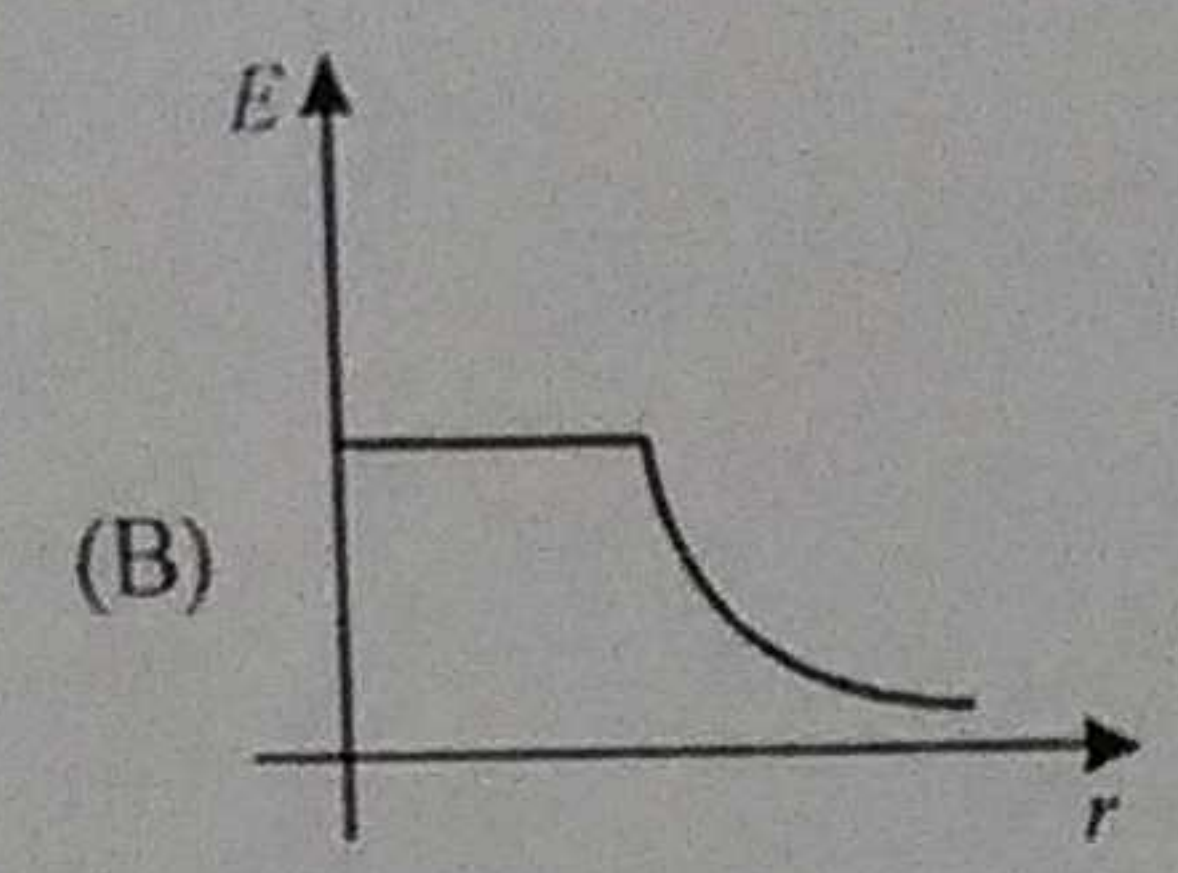
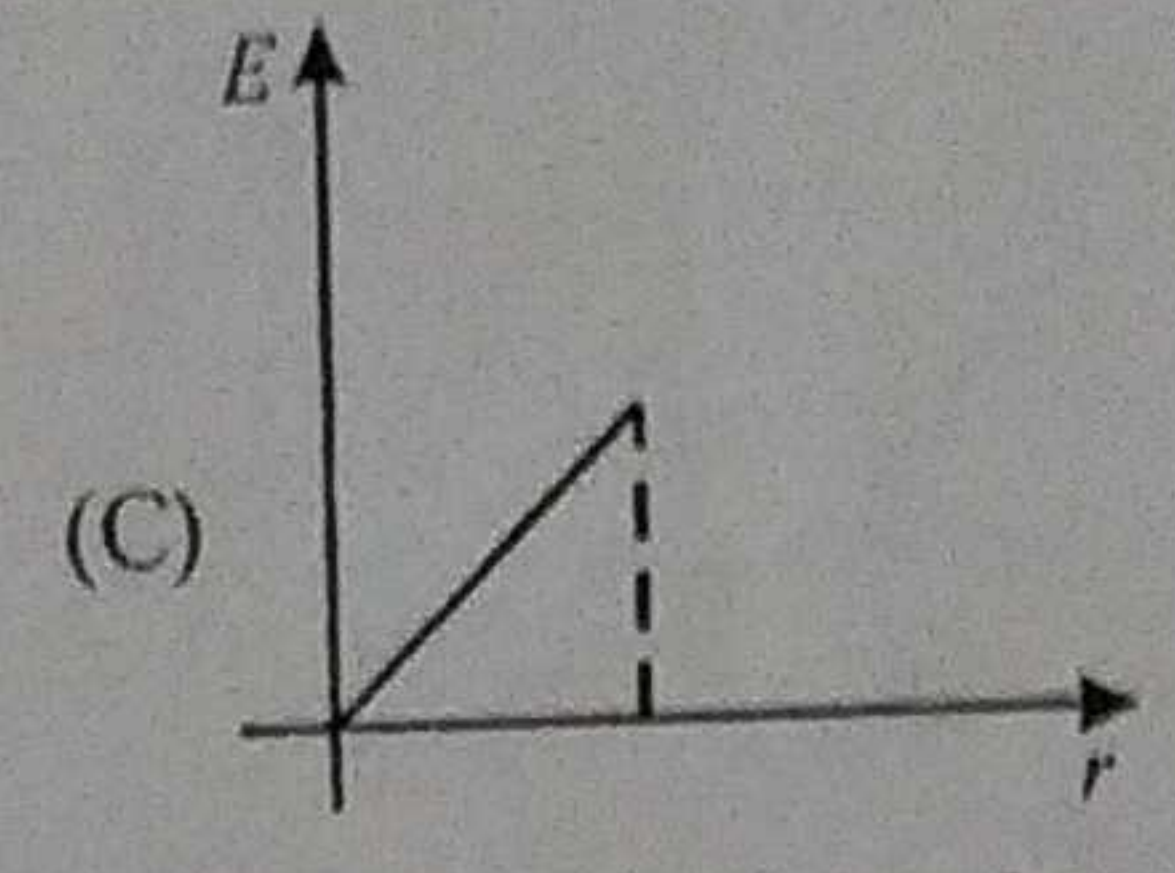
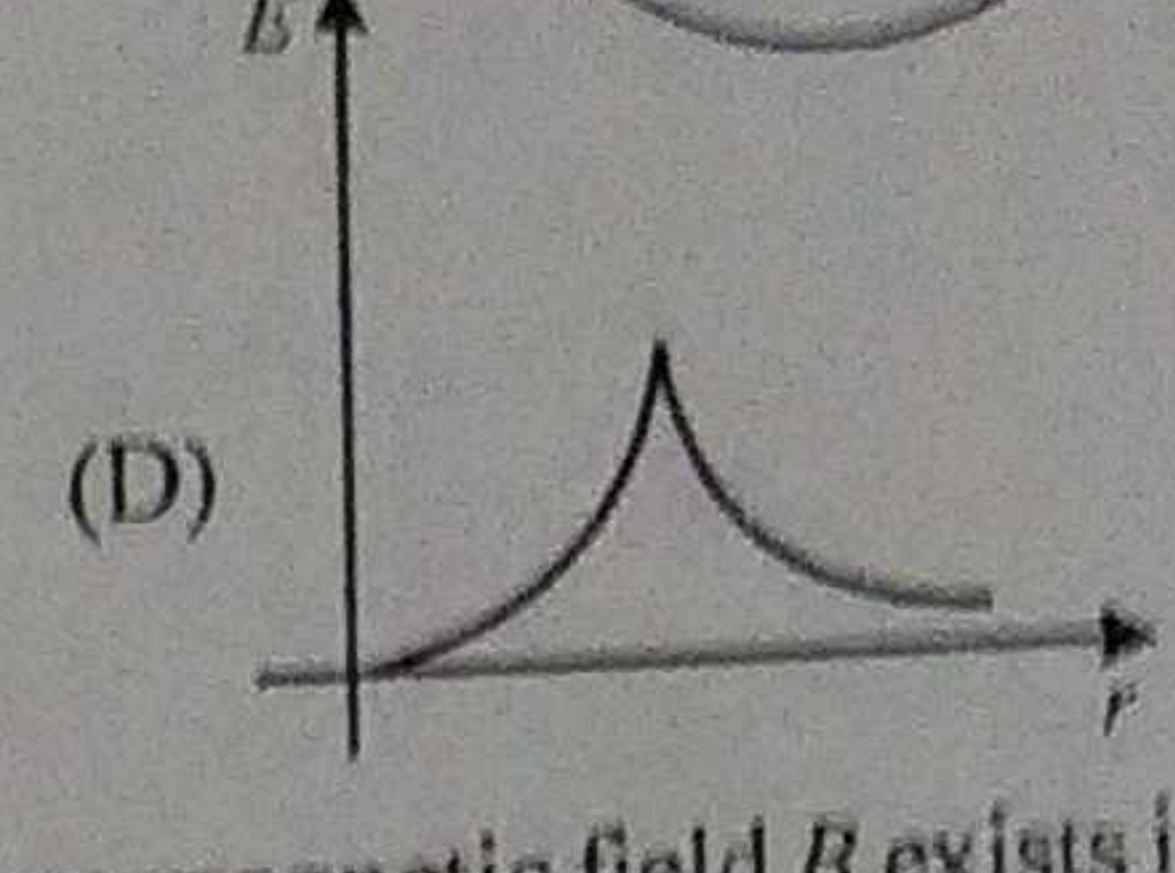
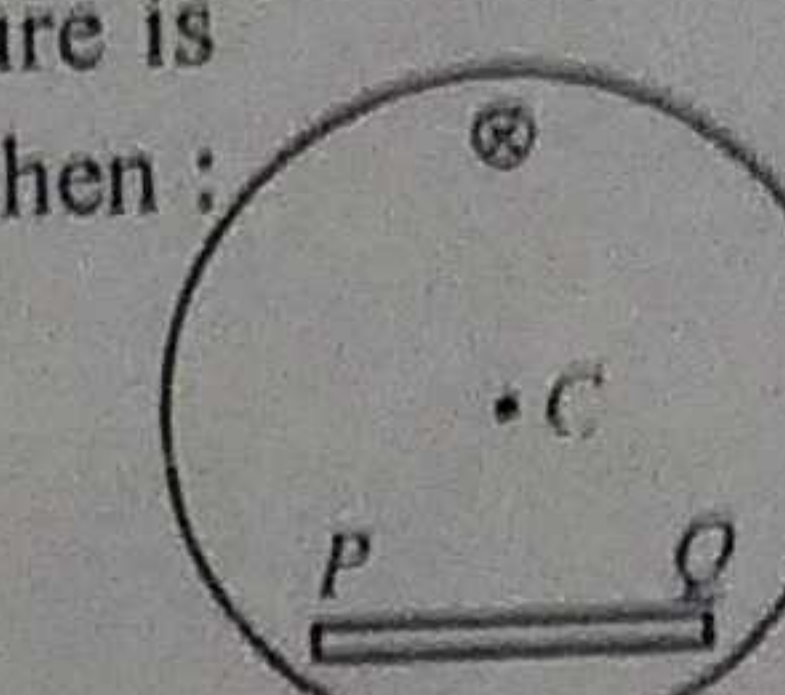
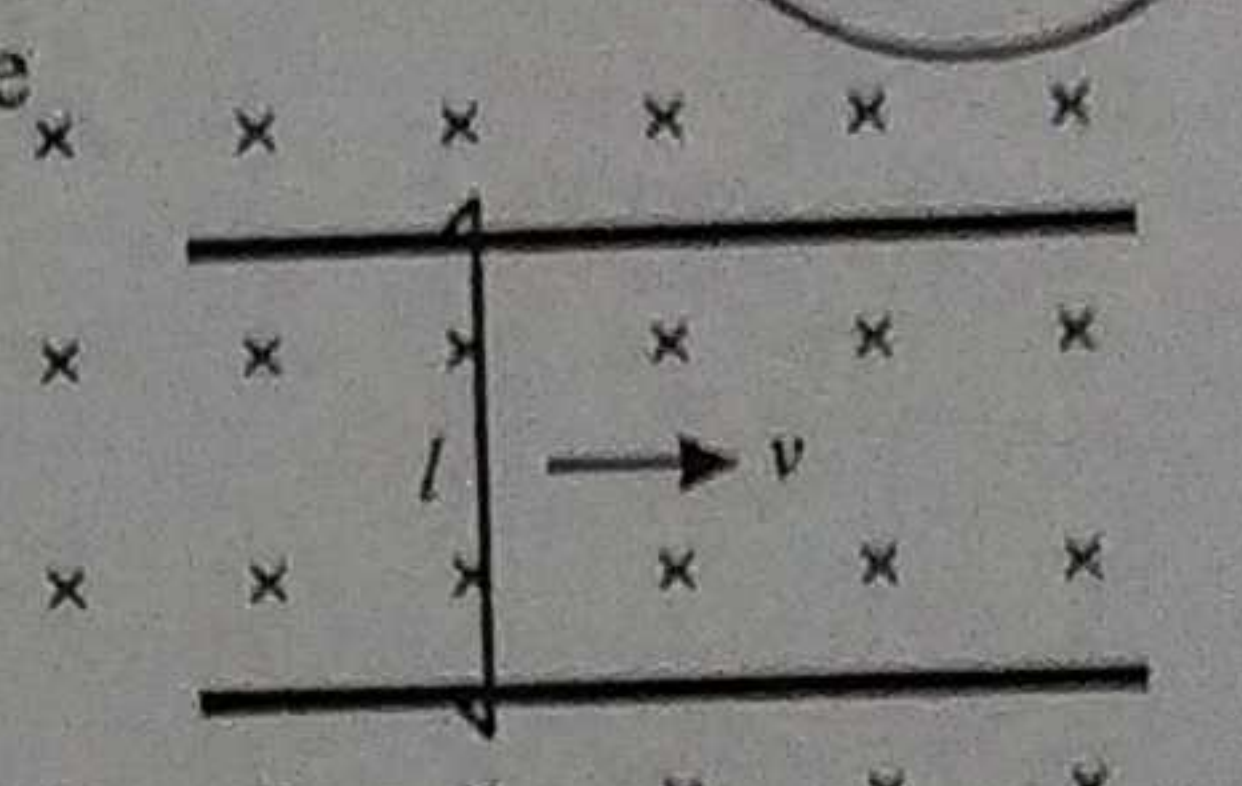
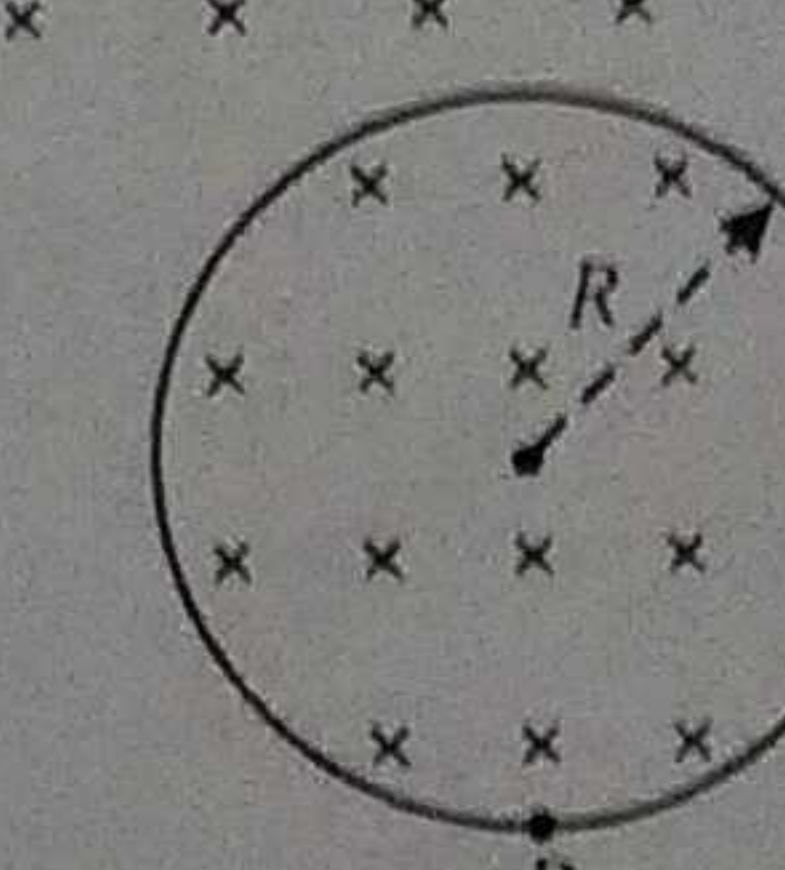
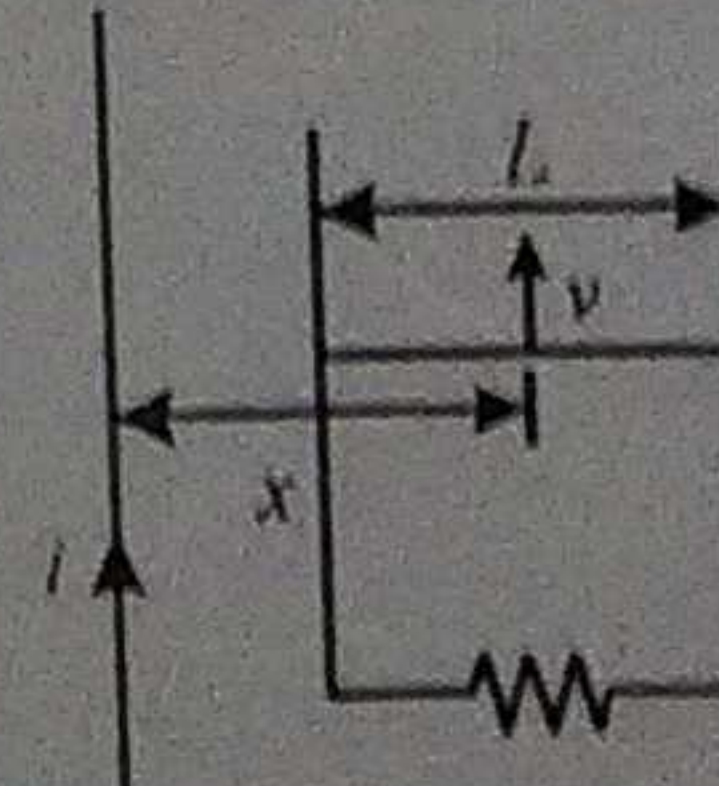
7. A wire of mass  $m$  and length  $l$  can slide freely on a pair of smooth, vertical rails (figure). A magnetic field  $B$  exists in the region in the direction perpendicular to the plane of the rails. The rails are connected at the top end by an initially uncharged capacitor of capacitance  $C$ . Find the acceleration of the wire neglecting any electric resistance.



8. A rod of length  $l$  rotates with a uniform angular velocity  $\omega$  about its perpendicular bisector. A uniform magnetic field  $B$  exists parallel to the axis of rotation. The potential difference between the two ends of the rod is :

- (A) Zero (B)  $\frac{1}{2} \omega B l^2$  (C)  $B \omega l^2$  (D)  $2B \omega l^2$



9. A wire of length 10 cm translates in a direction making an angle of  $60^\circ$  with its length. The plane of motion is perpendicular to a uniform magnetic field of 1.0 T that exists in the space. Find the emf induced between the ends of the rod if the speed of translation is 20 cm/s.
10. The two rails, separated by 1m, of a railway track are connected to a voltmeter. What will be the reading of the voltmeter when a train travels on the rails with speed 50 m/s. The earth's magnetic field at the place is  $0.4 \times 10^{-4}$  T, and the angle of dip is  $30^\circ$ .
11. A cylindrical space of radius  $R$  is filled with a uniform magnetic induction  $B$  parallel to the axis of the cylinder. If  $B$  changes at a constant rate, the graph showing the variation of induced electric field with distance  $r$  from the axis of cylinder is :
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- (A)  (B)  (C)  (D) 
12. A circular copper-ring of radius  $r$  translates in its plane with a constant velocity  $v$ . A uniform magnetic field  $B$  exists in the space in a direction perpendicular to the plane of the ring. Consider different pair of diametrically opposite points on the ring. (a) Between which pair of points is the emf maximum ? (b) Between which pair of points is the emf minimum ? What is the value of this minimum emf ?
13. In a cylindrical region uniform magnetic field which is perpendicular to the plane of the figure is increasing with time and a conducting rod  $PQ$  is placed in the region. If  $C$  is the centre of the circle then :
- (A)  $P$  will be at higher potential than  $Q$   
 (B)  $Q$  will be at higher potential than  $P$   
 (C) Both  $P$  and  $Q$  will be equipotential  
 (D) no emf will be developed across rod as it is not crossing / cutting any line of force
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14. Figure shows a wire sliding on two parallel, conducting rails placed at a separation  $l$ . A magnetic field  $B$  exists in a direction perpendicular to the plane of the rails. What force is necessary to keep the wire moving at a constant velocity  $v$  ?
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15. A uniform magnetic field of induction  $B$  is confined to a cylindrical region of radius  $R$ . The magnetic field is increasing at a constant rate of  $\frac{dB}{dt}$  (tesla/second). An electron of charge  $q$ , placed at the point  $P$  on the periphery of the field experiences an acceleration :
- (A)  $\frac{1}{2} \frac{eR}{m} \frac{dB}{dt}$  toward left (B)  $\frac{1}{2} \frac{eR}{m} \frac{dB}{dt}$  toward right  
 (C)  $\frac{eR}{m} \frac{dB}{dt}$  toward left (D) zero
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16. A conducting rod slides on a pair of thick metallic rails laid parallel to an infinitely long fixed wire carrying a constant current  $i$ . The center of the rod is at a distance  $x$  from the wire. The ends of the rails are connected by resistor of resistance  $R$ .
- (a) What force is needed to keep the rod sliding at a constant speed  $v$ , as shown in figure ?  
 (b) In this situation what is the current in the resistance  $R$  ?  
 (c) Find the rate of heat developed in the resistor.  
 (d) Find the power delivered by the external agent exerting the force on the rod.
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## ANSWER KEY

1. [Zero] 2. (D) 3. [(a) zero, (b)  $vB(bc)$ , positive at  $b$ , (c)  $vB(bc)$ , positive at  $a$ , (d) zero] 4. (C)  
 5.  $\left[ \frac{Blv}{2r(l+vt)} \right]$  6. [(a)  $\frac{B^2 l^2 v}{2r(l+vt)}$ , (b)  $l/v$ ] 7.  $\left[ \frac{mg}{m + CB^2 l^2} \right]$  8. (A) 9.  $[17 \times 10^{-3} \text{ V}]$  10.  $[1 \text{ mV}]$  11. (A)  
 12. [(a) at the ends of the diameter perpendicular to the velocity,  $2rvB$  (b) at the ends of the diameter parallel to the velocity, zero.] 13. (B) 14. [zero] 15. (A)  
 16. [(a)  $\frac{v}{R} \left( \frac{\mu_0 i}{2\pi} \ln \frac{2x+l}{2x-l} \right)^2$ , (b)  $\frac{\mu_0 i v}{2\pi R} \ln \frac{2x+l}{2x-l}$ , (c)  $\frac{1}{R} \left( \frac{\mu_0 i v}{2\pi} \ln \frac{2x+l}{2x-l} \right)^2$ , (d) same as (c)]